

Homework 4 (Part 0) - Real World Article / Chart Challenge

(25 points total)

Article Title	In America's Factories, Even the Robots Are Getting Less Work
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1. In class (Chapter 7 of Knafllic (2015), page 166), we learned that a good story “grabs your attention”, “takes you on a journey “, “evoking an emotional response”. Identify and discuss 5 paragraphs in the article that either grabbed your attention, took you on a journey or evoked an emotional response. In your answer to this question, put each of your selected paragraphs in quotation marks and lay out your discussion below the quoted paragraph. [5 points]

Focus of this question
• This question encourages you to evaluate the arguments, evidence, assumptions, and conclusions about key issues (i.e. think critically about the article) • This question encourages you to develop your own knowledge, comprehension and conceptual understanding and to connect, synthesize, and/or transform your ideas into a new form (i.e. be a creative thinker and contribute your ideas and thoughts)

Write your answer below:

“But Schron said some companies that bought robots during the pandemic-driver labor crunch underestimated the maintenance and programming skill needed to deploy them to more complicated tasks.”

The first thought I had after reading this paragraph was “oh, that makes sense”, and the emotion behind it was surprise and realization. The paragraphs before were mostly about why companies were buying robots (covid being one of the main reasons), and where business is stalling for some factory robotics companies. This paragraph introduces the reason why companies are not buying anymore. I could have guessed that the end of the pandemic would be a reason, but when the reason for maintenance was introduced, it was a reason that I hadn't considered/didn't think would impact the factory robotic industry that heavily.

“Robot manufacturers said customers have become choosier about their investments. High interest rates and lower production volumes mean that it is taking companies longer to recoup the money spent on robots.”

This paragraph also made me audibly say “oh, that makes sense, with satisfaction and realization being the emotion behind my reaction. Before reading this article and just looking at the headline, I was kind of confused as to why companies are not investing in robots more for warehouse tasks and other mundane jobs that could replace a human counterpart. Reading this paragraph answered the question, “how is replacing humans affecting companies on a financial level?” in a way that I did not expect. My assumption that I made was that replacing humans with robots would be more cost-effective, yet this article states otherwise. This paragraph leads me to believe that companies that have invested in robots are now struggling to put them to good use.

“Companies’ interest in automation is creating anxiety for some workers who see their jobs being threatened by robots, artificial intelligence and other technology. Dockworkers continue to negotiate limits on expanded use of automated equipment and vehicles at East and Gulf Coast ports after reaching a tentative agreement on wages with shipping companies last week, which ended the worker’s strike.”

This paragraph took me on a journey. It started as a problem where workers are worried about robots taking their jobs and ended with what is (hopefully) an agreement between the companies and their workers. I know I can empathize with people anxious about robots and AI taking over their jobs (the software industry is basically dead at this point), and this paragraph gave me hope to wish that these workers are being accommodated for by the shipping companies and that they can come to an agreement about what is best for both employer and employee.

“About 21% of manufacturers cited labor shortfalls as impediment to full production during the second quarter, down from 45% in the same quarter in 2022, according to Census Bureau survey data compiled by Michigan State University supply management professor Jason Miller. Material shortages were cited by less than 12% of manufacturers, compared with 39% two years earlier.”

This paragraph satiated my need for data-driven journalism. I know the reasons behind why companies aren’t buying robots anymore, but I need the data to back it up. This paragraph also took me on a journey of how companies are starting to see less employee turnover, and less material shortage (aka more resources). The emotion it evoked in me was hope that companies will start to invest more in humans rather than robot to have their production needs met especially now that more companies are seeing less turnover.

“Automakers and their suppliers had been buying more robots to outfit plants and increase electric-vehicle production. But U.S. auto companies have hit the brakes on some new EV models in response to disappointing sales.”

This paragraph grabbed my attention the most out of any other paragraph. In fact, it interested me so much, that I navigated away to both links provided in the article to investigate more. I know that EVs aren't the car of choice in the US, but revealing the statistics in the prior paragraph (that robot orders are down nearly 60%) made me realize how much larger the problem really is. It really surprised me (maybe from my lack of critical thinking), that because EVs aren't as popular, then that means production of EVs will go down, and thus the robots whose purpose it is to make specialized parts for these EVs are actually a really poor investment and make these companies (potentially) bleed money. It was surprising that a chain of events such as a decrease in EV sales could hurt robotic assembly lines.

2. Chapter 7 of Knaflig (2015), pages 175 - 177 talks about “The narrative structure” – and offers *two* ways to order a story. Read the article and classify the overall story in one of the two ways to order a story. Use arguments, including evidence from the article to justify your answer. [5 points]

Focus of this question
• This question encourages you to develop your own knowledge, comprehension and conceptual understanding and to connect, synthesize, and/or transform your ideas into a new form (i.e. be a creative thinker and contribute your ideas and thoughts) • This question encourages you to connect your learning to “real world” issues or life experiences and consider diverse perspectives for the application of concepts in the paper to the real world

Write your answer below:

In Chapter 7 of the textbook, the two different ways the authors offered to order a story are chronologically, and leading with the ending. This article does the latter and leads with the ending. The first sentence of the article is “Robots are getting less work at U.S. factories”. The audience is already given the outcome of the article even by way of the headline which reads “In America’s Factories, Even the Robots Are Getting Less Work”. The audience is already clued into what the article will be about, and what the outcome is. What the audience is here to do is to understand WHY robots are getting less work. There is some evidence of events being retold in chronological order such as the second paragraph which says, “orders for factory robots in North America plunged by nearly one-third last year from 2022’s record volume...” The audience is given reason as to how orders have dropped for factory robots, and in the continuing paragraph (or next page, rather) we are given a reason why which reads “Automation equipment attracted heavy investment after the Covid-19 pandemic left many factory jobs unstaffed and hard to fill...” The article goes on to list several other reasons why factory robots aren’t being ordered as much including “some companies that bought robots underestimated the maintenance and programming skill needed”, and “Dockworkers continue to negotiate limits on expanded use of automated equipment and vehicles at East and Gulf Coast ports after reaching a tentative agreement on wages.” As the article goes on, the audience gains more insight into the decrease of factory workers, which ultimately supports the very first sentence and headline of the article.

3. There is only one chart in the article. Discuss the overall message the chart sends. Identify/list at least two (2) stakeholders mentioned the article that the chart has implications for. Discuss the implications of the chart for each of your chosen stakeholders.

[15 points]

Focus of this question
• This question encourages you to connect your learning to “real world” issues or life experiences and consider diverse perspectives for the application of concepts in the article to the real world • This question encourages you to reflect on what you are learning • This question encourages you to contribute your ideas and thoughts • Consider diverse perspectives (gender, political, ethnic, racial, etc.) • Challenges you to develop and present your own knowledge, comprehension, and conceptual understanding

Write your answer below:

The sole chart in the article is one titled “Quarterly robot orders across all industries, North America”.

Two stakeholders that this chart has implications for is the Cleaveland automotive supplier Stripmatic Products which supplies robots for EV production, and Cleaveland company Jergens which makes robot arms.

For both companies demand in robots for factory assembly or anything else looks to be a lost cause as of now as the world adjusts back to a life without the caution of Covid-19 in the air anymore. In addition to in-person works starting up again, this implies that more humans are readily available for factory jobs again, and because data from Michigan State University has shown that there are lower rates of labor shortfalls (21% instead of 40%) as well as not as many material shortages (12% instead of 39%) which also decreases the need for robots in factories anymore.

Overall, the need for robots is lessening, and for Stripmatic Products and Jergens, this could mean a loss of profits over the coming years.